Mohammad Amin Ramezan Dehnavi

Machine Learning and Computer Vision Engineer

Tehran, Iran | +98 912 051 7266 | amindhnv@gmail.com | amin-dehnavi | amindehnavi

EDUCATION

M.Sc., Electrical Engineering (Digital Systems) | Sharif University of Technology *2020 – 2023*

- Thesis: Transformer-based multi-object tracking — improved MOT accuracy via a spatial-temporal aggregation block.
- Advisor: Dr. M. Hashemi.
- GPA: 17.68/20

B.Sc., Electrical Engineering (Electronics) | Iran University of Science and Technology *2016 – 2020*

- Thesis: YOLOv4-based pedestrian detection for mobile devices under low-light conditions.
- Advisor: Dr. S. Baradaran Shokouhi.
- GPA: 18.35/20

PROFESSIONAL EXPERIENCE

Deep Learning & Computer Vision Engineer | Part AI Research Center, Tehran *2023 – Present*

Iran's leading applied AI research center; developer of VIRA, an AI super application.

- **Diffusion-based object removal (VIRA Application).** Designed and built a diffusion inpainting model for object removal in cluttered scenes, released as a module of the image-editing suite in VIRA — Part AI's consumer AI application with 3.9M+ installs on Cafe Bazaar (Link). Raised removal success rate by **16%** and cut training time **71**× compared to the best open-source models at that time through custom architectural optimization. Work submitted to WACV 2027.
- **Production deployment and monitoring.** Packaged the object-removal model as a Docker container and integrated it with the platform's serving API, automating build, test, and release through GitLab CI. Tracked the live service in Grafana, monitoring resource utilization and service logs to diagnose failures in production.
- **CCTV object detection pipeline (Atlas Surveillance System).** Work on a multi-object tracking pipeline on NVIDIA DeepStream that serves 30 concurrent camera streams on a single RTX 3080 GPU. Reduced false-positive alerts by **10%**, which cut unnecessary surveillance notifications and improved operator efficiency. Work on further optimization of inference accuracy and throughput for production deployment is ongoing.
- **Face anonymization service (VIRA API marketplace).** Built a privacy-preserving face blurring model, released as a standalone service on Part AI's services marketplace.
- **Interactive background removal (VIRA API marketplace).** Built a background-removal service in which the user selects the object they want to keep and the model segments that object from the rest of the scene, returning it as a cut-out with the background stripped away so that it can be saved and reused in later edits.
- Recognized as **Part AI Best Computer Vision Engineer (2025)** across the research organization.

Research Assistant — Robust Vision Transformers | Sharif University of Technology *2022 – 2023*

- Integrated a sparse attention mechanism into Vision Transformer architectures to harden them against adversarial attacks; designed and ran the evaluation and ablation studies. Supervisor: Dr. S. Amini.

Research Assistant — High-Performance Edge ML | Sharif University of Technology *2020 – 2021*

- Designed and implemented machine learning algorithms for real-time inference on resource-constrained edge devices, working within explicit latency and memory budgets. Supervisor: Dr. M. A. Maddah-Ali.

Research Assistant — Feature Selection | Iran Univ. of Science and Technology *2019 – 2020*

- Benchmarked feature selection strategies for data mining and machine learning pipelines, comparing methods on predictive performance and dimensionality reduction. Supervisor: Prof. S. M. Mousavi Mirkalaei.

TECHNICAL SKILLS

Languages: Python, C/C++, CUDA, MATLAB

ML & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, NumPy, SciPy, Pandas

MLOps & Deployment: Docker, GitLab CI, Grafana, containerized model serving, production model monitoring

Modeling: Transformers and Vision Transformers, diffusion and generative models, object detection and multi-object tracking, CNN architectures, image inpainting, evaluation and ablation design

Data & Retrieval: FAISS vector search and similarity indexing, embedding-based retrieval

Systems: NVIDIA DeepStream framework, training-throughput optimization, edge deployment

SELECTED PROJECTS

- **Object Detection with Depth Estimation** (Project Page) — Model that detects objects and ranks them by distance from the camera; joint detection and monocular depth estimation.

- **Crowd Counting and Localization** (Project Page) — Density-based model estimating and localizing individuals in dense crowd scenes.
- **GPU-Accelerated Parallel Algorithms** — Implemented parallel algorithms and tiled matrix multiplication in CUDA, profiling memory access patterns against a CPU baseline.
- **Blood Pressure Estimation from PPG Signals** — Regression model estimating blood pressure from morphological features of photoplethysmogram waveforms; feature engineering on physiological time series.

PUBLICATIONS

- **Ramezan Dehnavi M. A.**, Fathi M., Akhlaghi A., Khanehgir S., Amini A. "Learning to Remove, Not Repeat: Robust Object Removal in Cluttered Scenes using Diffusion Models." Accepted for round 2 in WACV 2027.
- Mohaghegh S., **Ramezan Dehnavi M. A.**, Abdollahinejad G., Hashemi M. "PCGPT: Procedural Content Generation via Transformers." arXiv:2310.02405, 2023.
- **Ramezan Dehnavi M. A.**, Hamidi A., Baradaran Shokouhi S. "An optimal method for detecting pedestrians in hazy weather in self-driving cars using deep learning approach." AIRAFT Conference, Tehran, 2020. [Available on Civilica]

TEACHING & TECHNICAL COMMUNICATION

Invited Workshop Instructor — Multi-Modal AI | Hamrah Academy *Sept. 2026*

- Invited to deliver a workshop within the academy's 12-week Generative AI bootcamp. Hamrah Academy is the training arm of Hamrah-e-Aval (MCI), Iran's largest mobile network operator. (Website)

Invited Lecturer — "Transformers in Vision" | Sharif University of Technology *Nov. 2024*

- Delivered a technical talk on transformer architectures in computer vision to the graduate Deep Learning course.

Chief Teaching Assistant — Deep Learning | Sharif University of Technology *Fall 2022, 2023, 2024*

- Led a team of **15+ teaching assistants**, coordinating project design, grading standards, and tutorial delivery for a graduate-level course.

Teaching Assistant — Statistical Learning | Sharif University of Technology *Spring 2022, 2023*

- Ran scikit-learn and Python tutorial sessions, and designed and graded course projects.

HONORS

- **Part AI Best Computer Vision Engineer** *2025*
- **Direct-Ph.D. admission**, IUST Talented Student Office *2020*
  Also admitted to M.Sc. programs at Sharif, Tehran, and Amirkabir universities without entrance exam.
- **Ranked 1st of 130** students, Electrical Engineering department, IUST *2020*
- Top-3 student prize, four consecutive years *2016 – 2020*
- **Top 1%** of 162,000+ candidates, Iranian National University Entrance Exam (Mathematics and Physics) *2016*

LANGUAGES

Persian (native) | English (professional working proficiency, TOEFL iBT 100/120) | German (basic)

MILITARY SERVICE

Completed — holder of a Service Completion Card